

Optimization Methods in Machine Learning, CUB, Spring 2026

Theory 4. Conjugate functions and norms, proximal operators.

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1. Task 1

In all four cases $f^*(y) = \sup_x (xy - f(x))$ on the stated domain.

1.1. (a) $f(x) = x \log x - x$ on $[0, +\infty)$, $f(0) = 0$

For $x > 0$, $\partial_x(xy - x \log x + x) = y - \log x = 0 \implies x = e^y$. Then

$$f^*(y) = e^y y - e^y y + e^y = e^y.$$

The boundary value $x = 0$ gives $0 < e^y$, so

$$f^*(y) = e^y, \quad y \in \mathbb{R}.$$

1.2. (b) $f(x) = 1/x$ on $(0, +\infty)$

$$f^*(y) = \sup_{x>0} (xy - 1/x).$$

- $y > 0$: $xy - 1/x \rightarrow +\infty$ as $x \rightarrow \infty$.
- $y \leq 0$: $\partial_x = y + 1/x^2 = 0 \implies x = 1/\sqrt{-y}$ (for $y < 0$).

For $y < 0$:

$$f^*(y) = \frac{y}{\sqrt{-y}} - \sqrt{-y} = -\sqrt{-y} - \sqrt{-y} = -2\sqrt{-y}.$$

For $y = 0$: $\sup_{x>0} (-1/x) = 0 = -2\sqrt{0}$. Hence

$$f^*(y) = \begin{cases} -2\sqrt{-y} & y \leq 0 \\ +\infty & y > 0. \end{cases}$$

1.3. (c) $f(x) = -2\sqrt{-x}$ on $(-\infty, 0]$

Let $t = -x \geq 0$:

$$f^*(y) = \sup_{t \geq 0} (-ty + 2\sqrt{t}).$$

- $y \leq 0$: as $t \rightarrow \infty$, $-ty + 2\sqrt{t} \rightarrow +\infty$.
- $y > 0$: $\partial_t = -y + 1/\sqrt{t} = 0 \implies t = 1/y^2$, giving $-1/y + 2/y = 1/y$.

Hence

$$f^*(y) = \begin{cases} 1/y & y > 0 \\ +\infty & y \leq 0. \end{cases}$$

1.4. (d) $f(x) = \max(0, x)$ on $\mathbb{R}$

$$f^*(y) = \sup_x (xy - \max(0, x)).$$

- $x \geq 0$: $x(y - 1)$. Bounded above iff $y \leq 1$, then $\sup = 0$.

- $x \leq 0$: xy . Bounded above iff $y \geq 0$, then $\sup = 0$.

So

$$f^*(y) = \begin{cases} 0 & 0 \leq y \leq 1 \\ +\infty & \text{otherwise.} \end{cases}$$

2. Task 2

2.1. (a) $f(x) = \log \sum_{i=1}^n \exp(x_i)$
 $f^*(y) = \sup_x (x^T y - \log \sum_i \exp(x_i))$. Stationarity:

$$y_i = \frac{\exp(x_i)}{\sum_j \exp(x_j)} \implies y_i \geq 0, \quad \sum_i y_i = 1.$$

Then $x_i = \log y_i + \log Z$ with $Z = \sum_j \exp(x_j)$, and

$$x^T y - \log Z = \sum_i y_i \log y_i + \log Z \cdot \sum_i y_i - \log Z = \sum_i y_i \log y_i.$$

If y has $y_i < 0$ or $\sum_i y_i \neq 1$, shifting $x \rightarrow x + c\mathbf{1}$ shows the sup is $+\infty$. Let $\Delta_n = \{y \in \mathbb{R}^n : y \geq 0, \sum_i y_i = 1\}$. Therefore

$$f^*(y) = \begin{cases} \sum_{i=1}^n y_i \log y_i & y \in \Delta_n \\ +\infty & \text{otherwise} \end{cases}$$

with the convention $0 \log 0 = 0$.

2.2. (b) $f(x) = \max(x_1, \dots, x_n)$

Write $x_i = z + t_i$ with $t_i \leq 0$ and $\max_i t_i = 0$. Then $\max x_i = z$ and

$$x^T y - \max x_i = z \left(\sum_i y_i - 1 \right) + \sum_i t_i y_i.$$

Need $\sum_i y_i = 1$ (else $z \rightarrow \pm\infty$). With $t_i \leq 0$, $\sup \sum_i t_i y_i < \infty$ requires $y_i \geq 0$, and then $\sup = 0$ at $t = 0$. Hence

$$f^*(y) = \begin{cases} 0 & y \in \Delta_n \\ +\infty & \text{otherwise.} \end{cases}$$

2.3. (c) $f(X) = -\log \det X$ on S_{++}^n

$f^*(Y) = \sup_{X \succ 0} (\langle Y, X \rangle + \log \det X)$. Stationarity:

$$Y + X^{-1} = 0 \implies X = -Y^{-1}, \quad \text{requires } Y \prec 0.$$

Then

$$f^*(Y) = \langle Y, -Y^{-1} \rangle + \log \det(-Y^{-1}) = -n - \log \det(-Y).$$

For Y not in $-S_{++}^n$, the sup is $+\infty$. So

$$f^*(Y) = \begin{cases} -n - \log \det(-Y) & Y \prec 0 \\ +\infty & \text{otherwise.} \end{cases}$$

3. Task 3

3.1. (a) $f(x) = \|x\|_\infty$

Dual norm of $\|\cdot\|_\infty$ is $\|\cdot\|_1$. For any norm,

$$\partial\|x\| = \{y : \|y\|_* \leq 1, \langle y, x \rangle = \|x\|\}.$$

Let $I = \arg \max_i |x_i|$. If $x \neq 0$,

$$\partial f(x) = \text{conv}\{\text{sign}(x_i)e_i : i \in I\} = \left\{ y : \text{supp}(y) \subseteq I, y_i \text{sign}(x_i) \geq 0, \sum_{i \in I} y_i \text{sign}(x_i) = 1 \right\}.$$

If $x = 0$,

$$\partial f(0) = \{y : \|y\|_1 \leq 1\}.$$

3.2. (b) $f(x) = \sum_{1 \leq i < j \leq n} |x_i - x_j|$

By the sum rule, $\partial f(x) = \sum_{i < j} \partial |x_i - x_j|$, where

$$\partial |x_i - x_j| (x) = \begin{cases} \{e_i - e_j\} & x_i > x_j \\ \{e_j - e_i\} & x_i < x_j \\ \{s(e_i - e_j) : s \in [-1, 1]\} & x_i = x_j. \end{cases}$$

Equivalently, with $\text{Sgn}(t) = \{\text{sign}(t)\}$ for $t \neq 0$ and $[-1, 1]$ for $t = 0$:

$$\partial f(x) = \left\{ \sum_{i < j} s_{ij}(e_i - e_j) : s_{ij} \in \text{Sgn}(x_i - x_j) \right\}.$$

4. Task 4

Let $A = U\Sigma V^T$ be the SVD with $\Sigma = \text{diag}(\sigma_1, \dots, \sigma_n)$, $\sigma_i \geq 0$.

4.1. (a) Reduction to diagonal matrices

Set $Y = U^T X V$. Since U, V are orthogonal, $\|Y\|_2 = \|X\|_2$. Also

$$\langle A, X \rangle = \text{tr}(V\Sigma U^T X) = \text{tr}(\Sigma U^T X V) = \text{tr}(\Sigma Y) = \sum_{i=1}^n \sigma_i Y_{ii}.$$

The objective depends only on $\text{diag}(Y)$. For any Y , $\|Y\|_2 \geq \max_i |Y_{ii}|$, with equality at $Y = \text{diag}(Y_{11}, \dots, Y_{nn})$. Hence the optimum is attained at diagonal Y :

$$\|A\|_{2*} = \max_{d \in \mathbb{R}^n} \left\{ \sum_{i=1}^n \sigma_i d_i : \|\text{diag}(d)\|_2 = \|d\|_\infty \leq 1 \right\}.$$

4.2. (b)

Since $\sigma_i \geq 0$, the maximum is at $d_i = 1$ for all i :

$$\|A\|_{2*} = \sum_{i=1}^n \sigma_i(A),$$

attained at $Y = I_n$, i.e. $X^* = UV^T$.

5. Task 5

Let f be closed convex.

(a) $\implies$ (b). By definition $u = \arg \min_v (f(v) + \frac{1}{2} \|v - x\|^2)$. The optimality condition for a sum of a closed convex function and a smooth strictly convex one is

$$0 \in \partial f(u) + (u - x) \implies x - u \in \partial f(u).$$

(b) $\Leftrightarrow$ (c). Definition of subgradient:

$$g \in \partial f(u) \iff f(v) \geq f(u) + \langle g, v - u \rangle \quad \forall v.$$

Setting $g = x - u$ gives (c).

(c) $\implies$ (a). From (c), for any v ,

$$f(v) + \frac{1}{2} \|v - x\|^2 \geq f(u) + \langle x - u, v - u \rangle + \frac{1}{2} \|v - x\|^2.$$

Using $\|v - x\|^2 = \|v - u\|^2 + 2\langle v - u, u - x \rangle + \|u - x\|^2$,

$$f(v) + \frac{1}{2} \|v - x\|^2 \geq f(u) + \frac{1}{2} \|u - x\|^2 + \frac{1}{2} \|v - u\|^2 \geq f(u) + \frac{1}{2} \|u - x\|^2.$$

Hence u minimizes $f(v) + \frac{1}{2} \|v - x\|^2$, i.e. $u \in \text{prox}_f(x)$.

Thus (a) $\Leftrightarrow$ (b) $\Leftrightarrow$ (c).

6. Task 6

Let $u = \text{prox}_f(x)$, $v = \text{prox}_f(y)$. By Task 5(c),

$$f(v) \geq f(u) + \langle x - u, v - u \rangle,$$

$$f(u) \geq f(v) + \langle y - v, u - v \rangle.$$

Adding:

$$0 \geq \langle x - u, v - u \rangle + \langle y - v, u - v \rangle = \langle (x - y) - (u - v), v - u \rangle.$$

Hence

$$\langle u - v, x - y \rangle \geq \|u - v\|^2,$$

i.e.

$$\|\text{prox}_f(x) - \text{prox}_f(y)\|^2 \leq \langle \text{prox}_f(x) - \text{prox}_f(y), x - y \rangle. \quad (*)$$

By Cauchy-Schwarz, $\langle u - v, x - y \rangle \leq \|u - v\| \|x - y\|$, so from (*):

$$\|u - v\|^2 \leq \|u - v\| \|x - y\| \implies \|u - v\| \leq \|x - y\|.$$

Thus prox_f is a contraction (in fact, firmly non-expansive).